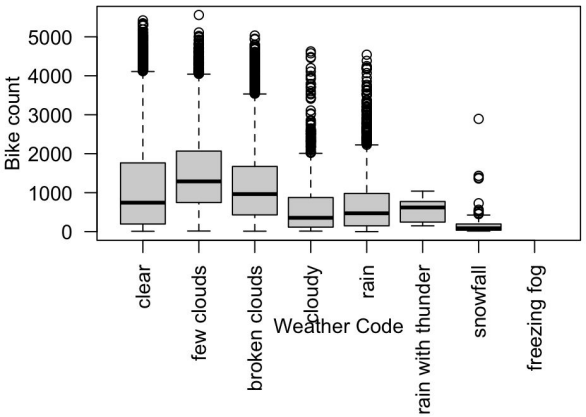
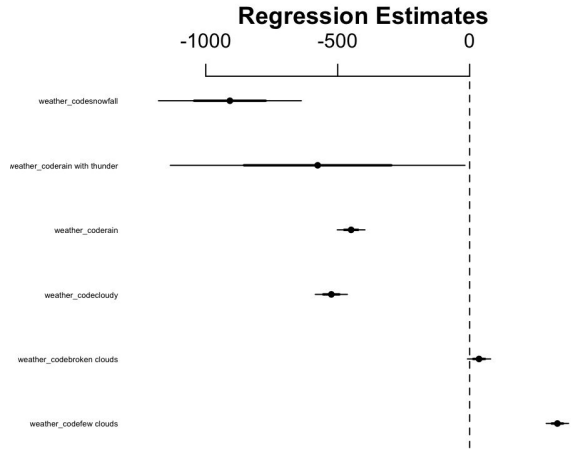
# Mini Analysis 2

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Wilhelm Friberg

# Categorical covariates - weather code

```
Coefficients:
                            Estimate Std. Error t value Pr(>|t|)
(Intercept)                  1159.27      13.30  87.132  < 2e-16 ***
weather_codefew clouds        332.78      21.14  15.741  < 2e-16 ***
weather_codebroken clouds      35.85      21.99   1.631   0.1030
weather_codecloudy           -524.04      30.34 -17.275  < 2e-16 ***
weather_coderain             -449.05      26.18 -17.151  < 2e-16 ***
weather_coderain with thunder -575.84     279.10  -2.063   0.0391 *
weather_codesnowfall          -908.42     135.32  -6.713 1.97e-11 ***
```

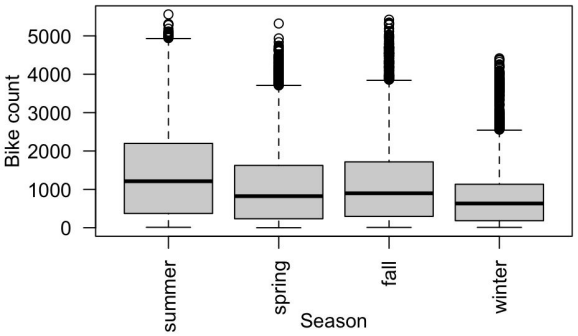


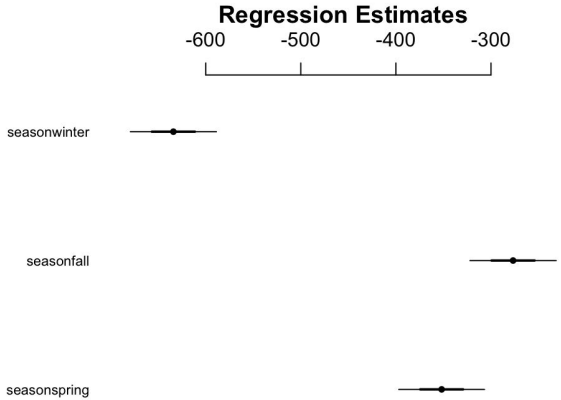




- broken_clouds not significantly different from baseline (clear) → Null hypothesis can't be rejected
- Generally expected interpretations, the "worse" weather, the less the average amount of rented bikes are compared to "clear" weather.

# Categorical covariates - season

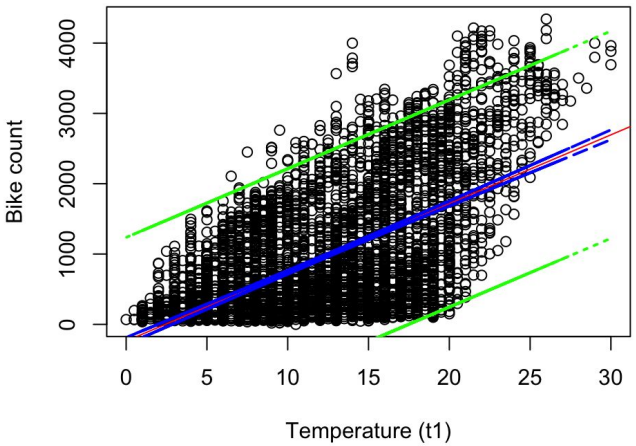
```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)   1455.73      15.94   91.30   <2e-16 ***
seasonspring  -351.90      22.53  -15.62   <2e-16 ***
seasonfall    -276.77      22.65  -12.22   <2e-16 ***
seasonwinter  -634.00      22.61  -28.04   <2e-16 ***
```




- All seasons have significant difference compared to baseline (summer)
- Again, expected interpretations with least bikes rented in winter

# Confidence intervals - Temperature




- Confidence interval - 95% of the time the mean predicted bike count is between the blue lines
  - Big data set → narrow spread
- Prediction interval - 95% of future individual observations for given temperature is between the green lines

# Multi Regression (t1 and hour of day)

```
Call:
lm(formula = data$cnt ~ data$t1 + data$hour_of_day)

Residuals:
     Min       1Q   Median       3Q      Max
-1613.96  -543.57   -89.05   467.50  2787.51

Coefficients:
                Estimate Std. Error t value Pr(>|t|)
(Intercept)     -570.510     27.951  -20.41   <2e-16 ***
data$t1           90.024      1.840   48.94   <2e-16 ***
data$hour_of_day  37.262      1.471   25.33   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 706.8 on 4967 degrees of freedom
Multiple R-squared:  0.4167,    Adjusted R-squared:  0.4164
F-statistic:  1774 on 2 and 4967 DF,  p-value: < 2.2e-16
```

```
Call:
lm(formula = data$cnt ~ data$hour_of_day)

Residuals:
    Min      1Q  Median      3Q     Max
-1460.1  -558.2  -242.8   384.5  3291.3

Coefficients:
                Estimate Std. Error t value Pr(>|t|)
(Intercept)      409.441     23.740   17.25   <2e-16 ***
data$hour_of_day  49.248      1.766   27.89   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 860.4 on 4968 degrees of freedom
Multiple R-squared:  0.1354,    Adjusted R-squared:  0.1352
F-statistic:   778 on 1 and 4968 DF,  p-value: < 2.2e-16
```

```
Call:
lm(formula = data$cnt ~ data$t1)

Residuals:
    Min      1Q  Median      3Q     Max
-1568.5  -566.1  -112.2   524.6  2867.2

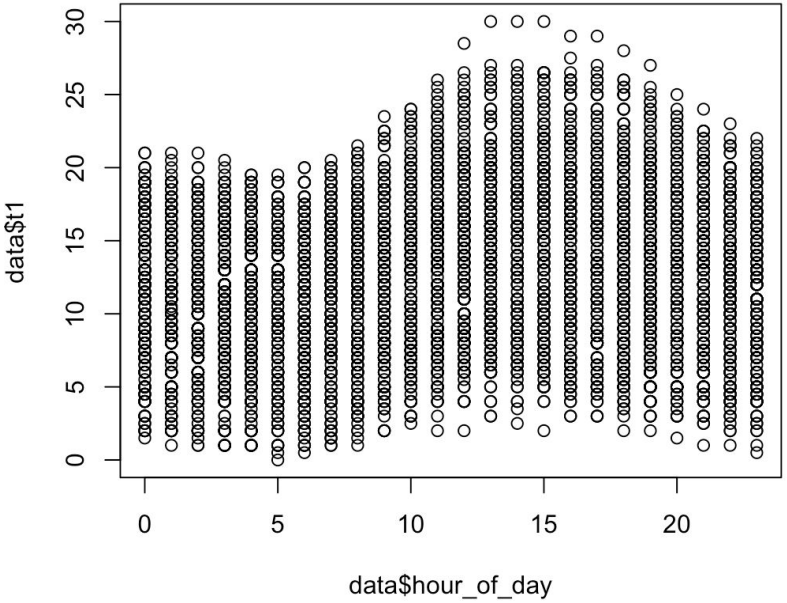
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -237.150     26.202  -9.051   <2e-16 ***
data$t1       97.783      1.927  50.736   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

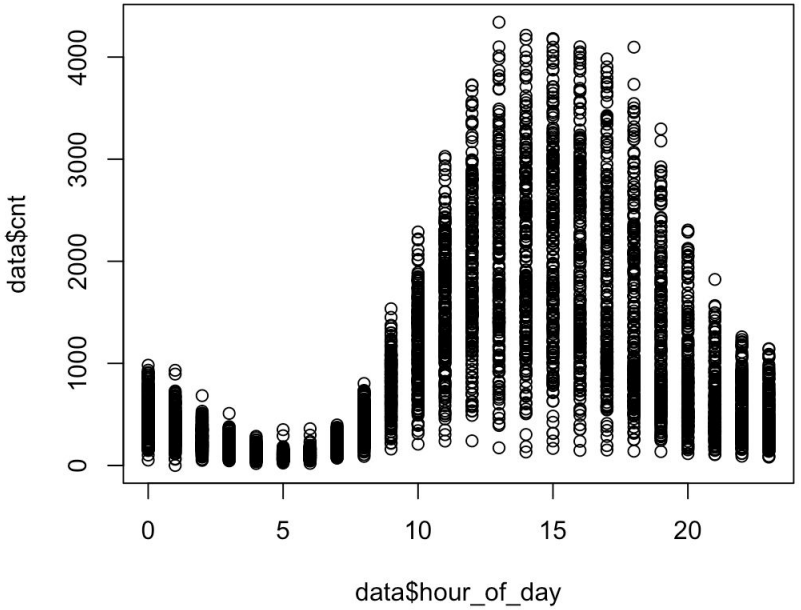
Residual standard error: 751 on 4968 degrees of freedom
Multiple R-squared:  0.3413,    Adjusted R-squared:  0.3412
F-statistic:  2574 on 1 and 4968 DF,  p-value: < 2.2e-16
```

Subset of data on weekend with count < 6000.

Utilizing a linear regression model with both t1 and hour of day as coverients $R^2 = 0.42$ (Not great correlation). However using t1 and hour of day separately is worse. T1 changes cyclically with time, as does cnt.

# (T1 vs hour of day) and (cnt vs hour of day)




Cyclical pattern with t1 and cnt vs hour of day. When temperature is high, bike count is high. When temperature is low, bike count is low

# Partial F test

```
Analysis of Variance Table

Model 1: data$cnt ~ data$t1 + data$hour_of_day
Model 2: data$cnt ~ data$t1 + data$hour_of_day + data$hum
  Res.Df        RSS Df Sum of Sq      F    Pr(>F)
1   4967 2481186919
2   4966 1688532343  1 792654576 2331.2 < 2.2e-16 ***
```

```
lm(formula = data$cnt ~ data$t1 + data$hour_of_day + data$hum)

Residuals:
     Min       1Q   Median       3Q      Max
-1887.64  -411.60   -42.82   364.72  3012.97

Coefficients:
                 Estimate Std. Error t value Pr(>|t|)
(Intercept)      2447.2905    66.6213   36.73   <2e-16 ***
data$t1            52.9522     1.7008   31.13   <2e-16 ***
data$hour_of_day   23.3920     1.2471   18.76   <2e-16 ***
data$hum          -32.8581     0.6805  -48.28   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 583.1 on 4966 degrees of freedom
Multiple R-squared:  0.603,     Adjusted R-squared:  0.6028
F-statistic:  2515 on 3 and 4966 DF,  p-value: < 2.2e-16
```

Partial F test of previous multiregression (T1 and hour of day) vs new multiregression (T1, hour of day and humidity). F test indicates that adding hum as coverient would improve the linear regression model. ($R^2$ = 0.603 with hum added).

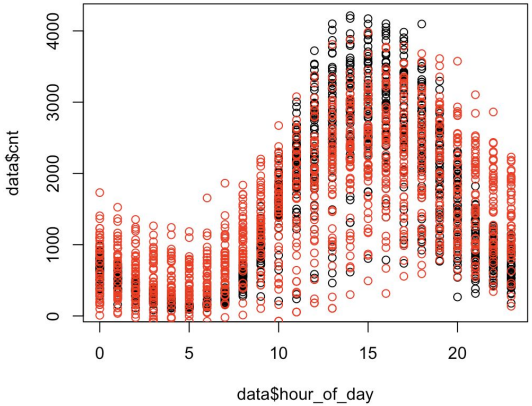
Use Model 2 to create fitted bike count.

VIF check of the coverients use (All <2, good non linear dependency)

```
> vif(l4)
       data$t1 data$hour_of_day          data$hum
      1.870023         1.232794         1.804492
```

# Fitted vs Predicted values

```
data$fitted <- l4$fitted.values
```

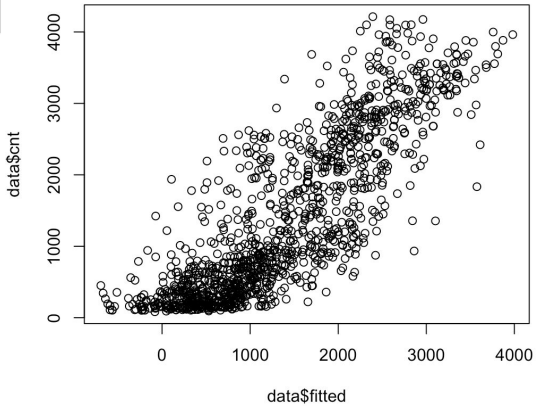



```
Call:
lm(formula = data$cnt ~ data$fitted)

Residuals:
     Min       1Q   Median       3Q      Max
-1928.40  -416.44   -55.39   354.76  1987.22

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 1.030e-12  3.073e+01    0.00        1
data$fitted 1.000e+00  1.831e-02   54.63   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

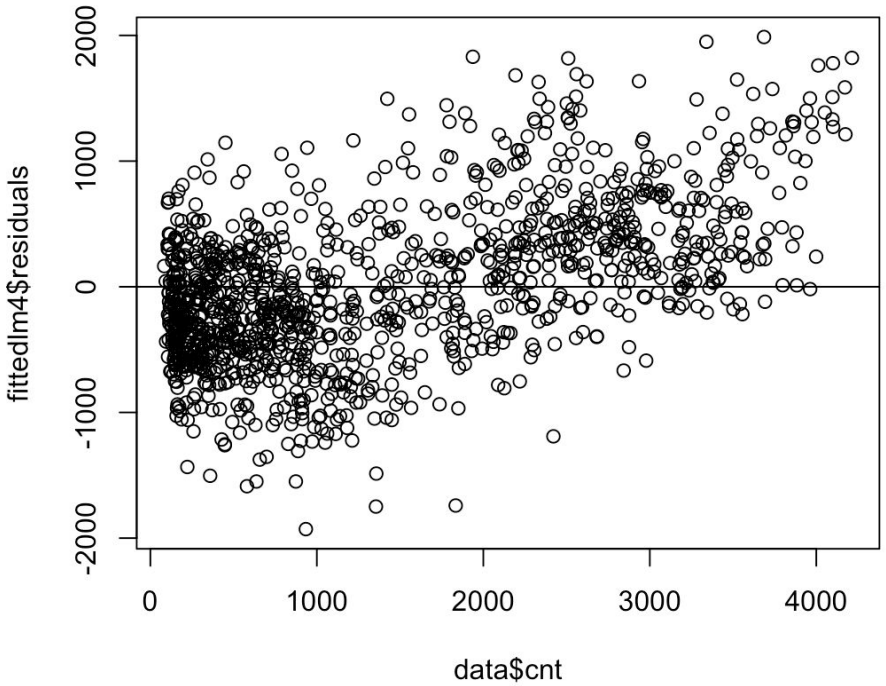
Residual standard error: 612.4 on 1245 degrees of freedom
Multiple R-squared:  0.7056,    Adjusted R-squared:  0.7054
F-statistic:  2984 on 1 and 1245 DF,  p-value: < 2.2e-16
```




Utilizing fitted.values function on the regression model in previous slide to create fitted values for the bike count (RED circles) and plotted with bike cnt over the hour of the day. Pretty good fitted values.

Plotting the bike cnt vs fitted values a nice linear regression is visible. (Some negative fitted values)

# Residuals for fitted count




Residuals of regression of fitted data vs predicted data. At low bike counts the model tends to overvalue while at higher bike counts it tends to undervalue the amount of bikes.

# Diagnostic of linear regression model

Q-Q residuals: Shows how well the residuales are gaussian distributed. More “linear” is closer to gaussian.

Scale-Location: Shows how scattered the residuals are, red line should be horizontal for full homoscedasticity (even scattering of residuals)

Cook’s distance: Shows how influential each data point is in the linear regression model

